Lot 37 — Moradia Unifamiliar Isolada

Urbanização do Espartal · Aljezur · Costa Vicentina

Architectural Concept — Design Development

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Lot 37 · Moradia Unifamiliar Isolada · Groenvastbouw

— Project Vision

A modern Portuguese villa where Algarve tradition meets precision engineering

This architectural concept presents a detaıched 4-suite residence on Lot 37 within Urbanização do Espartal, Aljezur. The house features a highly efficient, regular rectangular footprint (~142 m²). The hybrid construction system utilizes a solid concrete base that extends into the pool terrace, supporting two upper storeys built entirely with HSB (Hout Skelet Bouw) prefabricated timber frame elements. This ensures rapid assembly, superior thermal performance, and a clean, modern aesthetic.

PLOT AREA	634.05 m² (Irregular)
HOUSE FOOTPRINT	13.5m x 10.5m (Regular Rectangle)
BEDROOMS	4 suites
FLOORS	2 HSB storeys + 1 Concrete ground/cave
CONSTRUCTION	Concrete Base + HSB Timber Frame + Glulam Beams
EXTERIOR	White render + wood cladding

Site Context and Orientation

South-facing road, ocean views to the west — perfectly positioned

Lot 37 (634 m²) sits within Urbanização do Espartal with the access road running along its **SOUTH** boundary. The Atlantic Ocean lies to the **WEST**. The terrain slopes naturally from **+71 m** in the north down to **+67 m** at the road (south), providing a **4-meter** drop for a semi-embedded lower level.

ORIENTATION STRATEGY

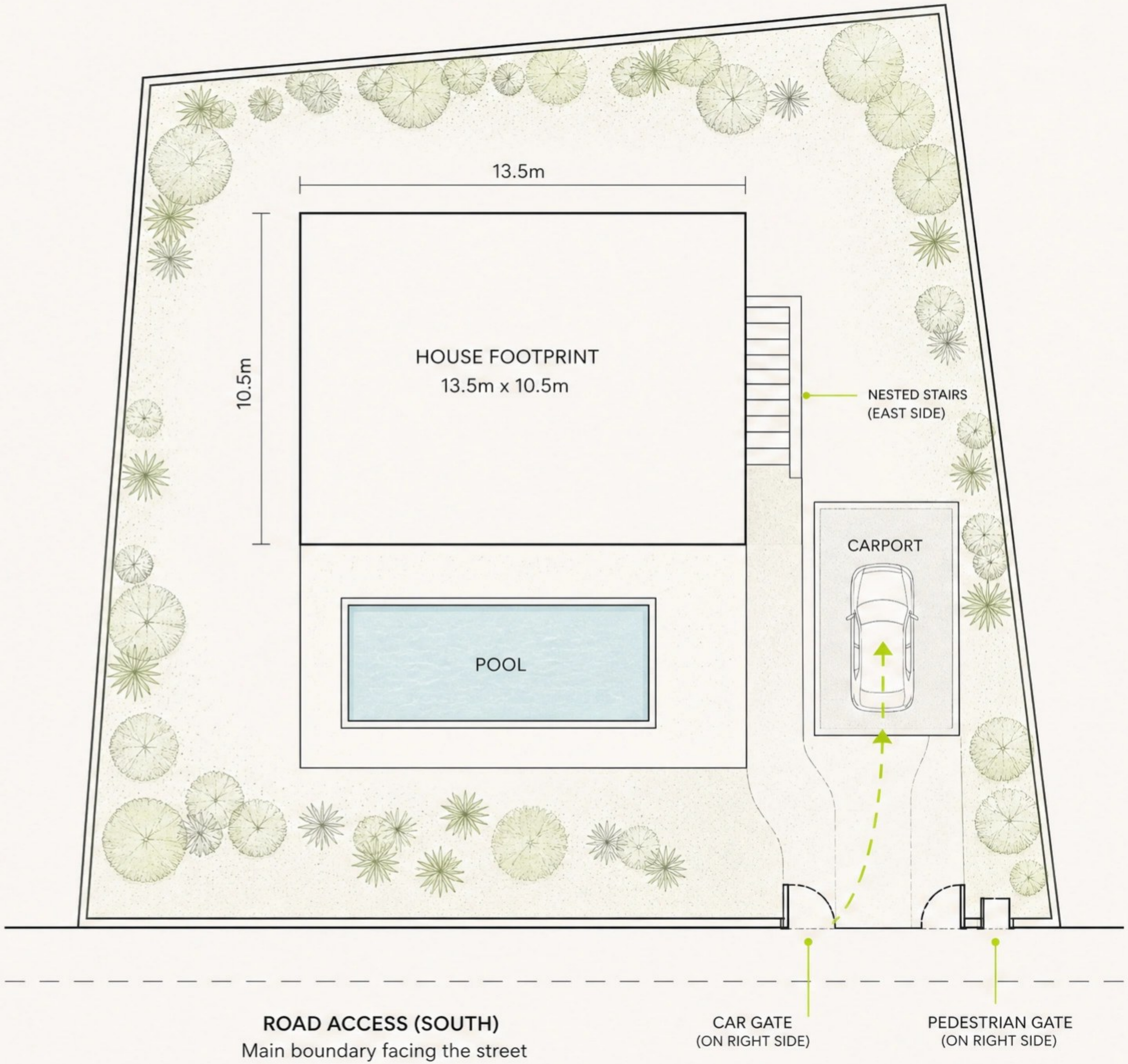
SOUTH:	Road access, pool, carport, main entrance
WEST:	Ocean views (living room & bedrooms)
NORTH:	Higher terrain, garden, privacy
EAST:	Morning sun, secondary bedrooms

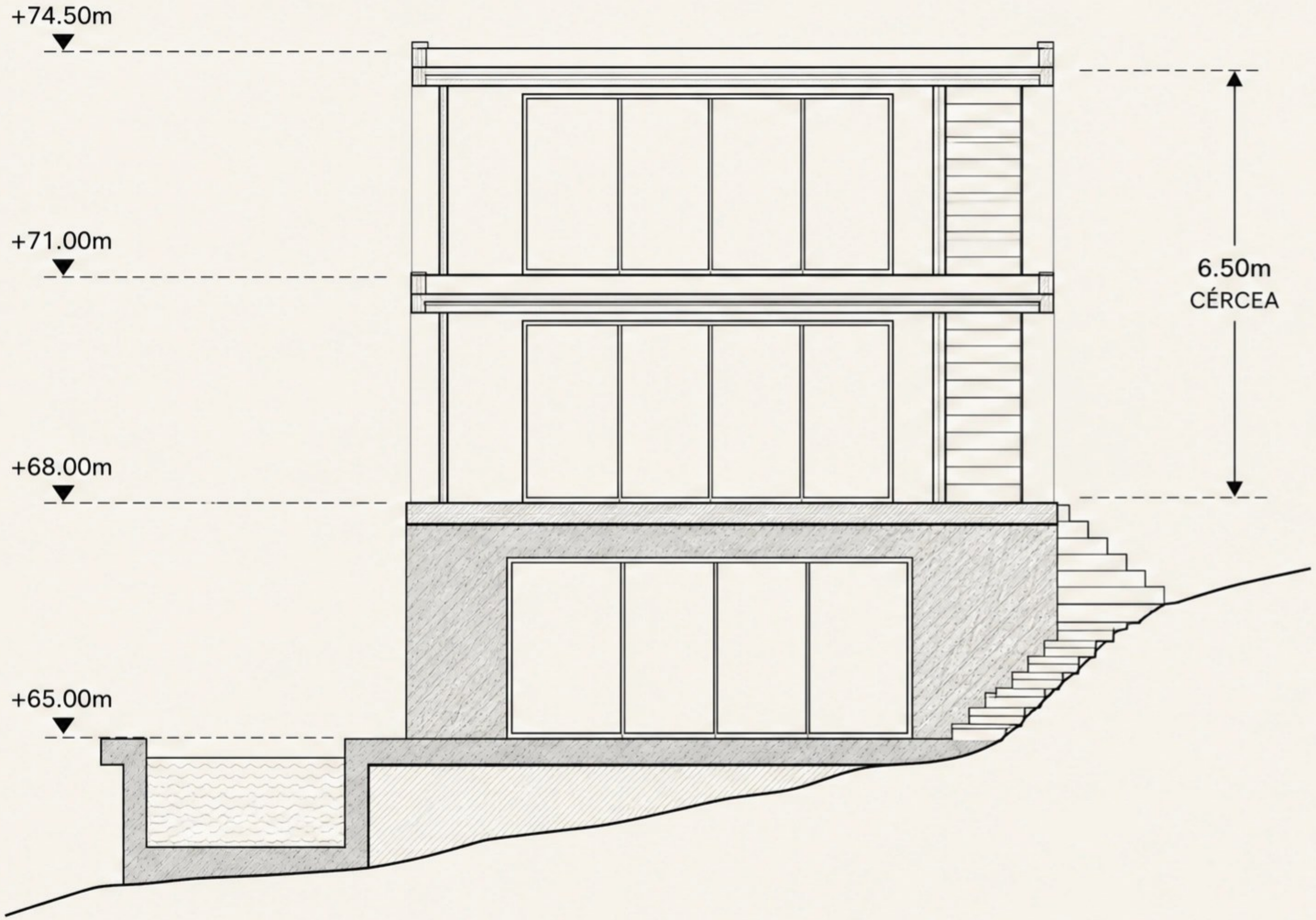
Site Plan — Top View

A regular geometric volume placed within an irregular landscape

→ **KEY POINTS (ENTRANCE & FLOW)**

- **Road Access (South):** Main boundary facing the street
- **Entrance Gates:** Car gate and pedestrian gate located on the **RIGHT SIDE (southeast corner)**
- **Carport:** Positioned immediately inside the car gate on the right side
- **House Footprint:** A strict 13.5m x 10.5m rectangle
- **Pool:** A perfectly rectangular pool on the south concrete slab
- **Nested Stairs:** Exterior concrete stairs nested on the east side of the south facade





Vertical Organization — Section

Concrete base supporting two lightweight HSB storeys with a flat roof

The structural logic is clearly divided. A reinforced concrete ground level sits partially embedded in the gentle slope, extending south to form the pool deck. Above this solid base, two storeys are constructed entirely from prefabricated HSB timber elements, capped by a perfectly flat roof.

LEVEL	ELEVATION	NOTES
RF Flat Roof	+74.50m	6.50m above soleira (At legal cércea limit)
02 Upper Floor (HSB)	+71.00m	4 Suites · 2.70m panel height
01 Main Floor (HSB)	+68.00m	Cota de Soleira · 2.70m panel height
G Ground/Gym (Concrete)	+65.00m	Below soleira (Does not count to cércea)

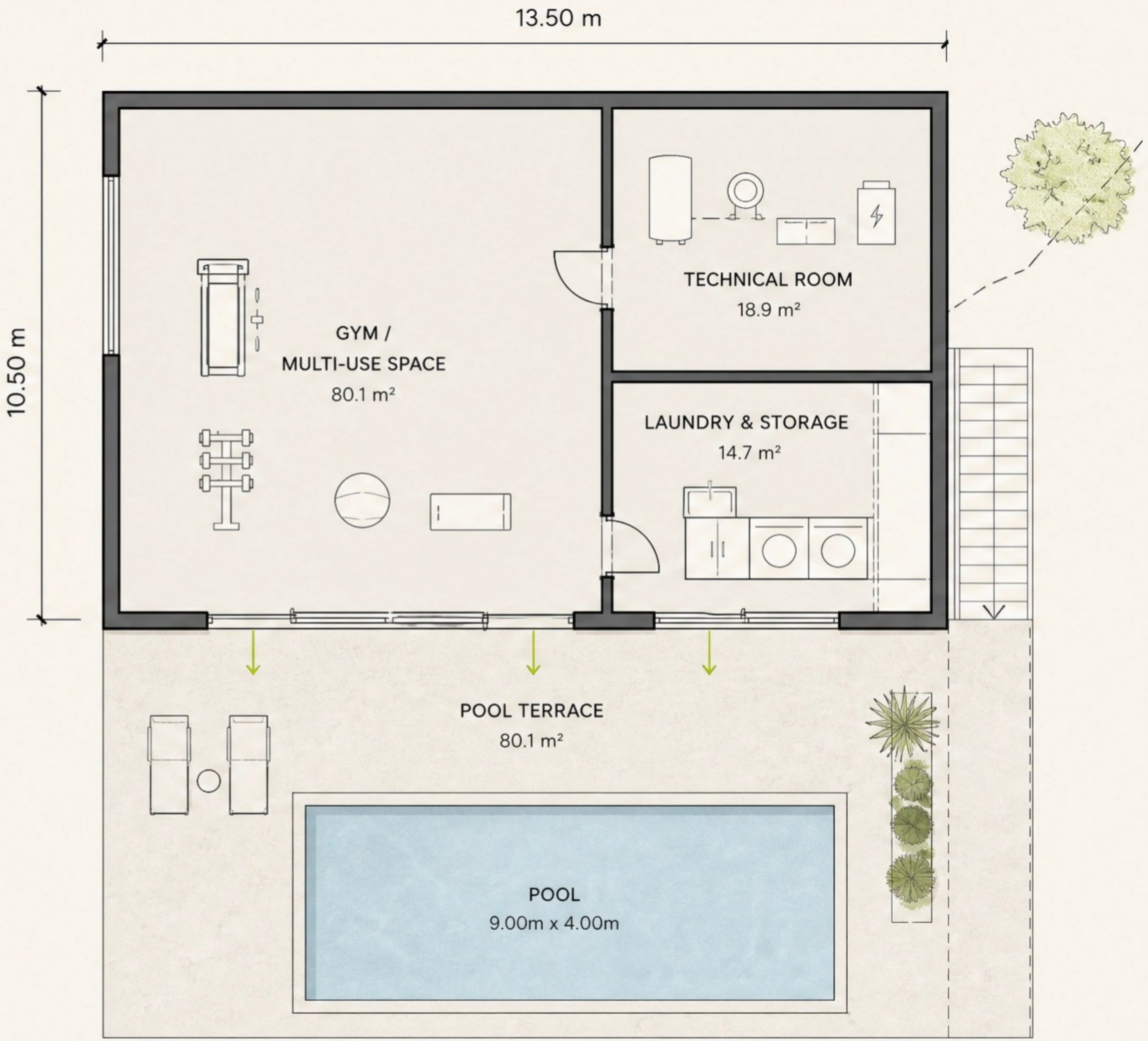
Ground Level — Cave & Pool Terrace

Solid concrete base embedded in the terrain

The ground floor is constructed entirely of reinforced concrete. Because of the gentle slope, the north side is partially embedded in the terrain, while the south side opens fully to the pool terrace. This level provides a solid foundation for the timber frame structure above.

KEY POINTS (PROGRAM & STRUCTURE)

- **Regular Rectangle:** Strict 13.5m x 10.5m footprint
- **Gym / Multi-use Space:** Large open area with natural light facing south
- **Technical Room:** Houses pool equipment, HVAC, and solar battery storage
- **Laundry & Storage:** Generous utility spaces
- **Pool Terrace:** The concrete slab extends south to form the deck around the rectangular pool
- **Ceiling Height:** Full 3.00m height (sits below the cota de soleira)

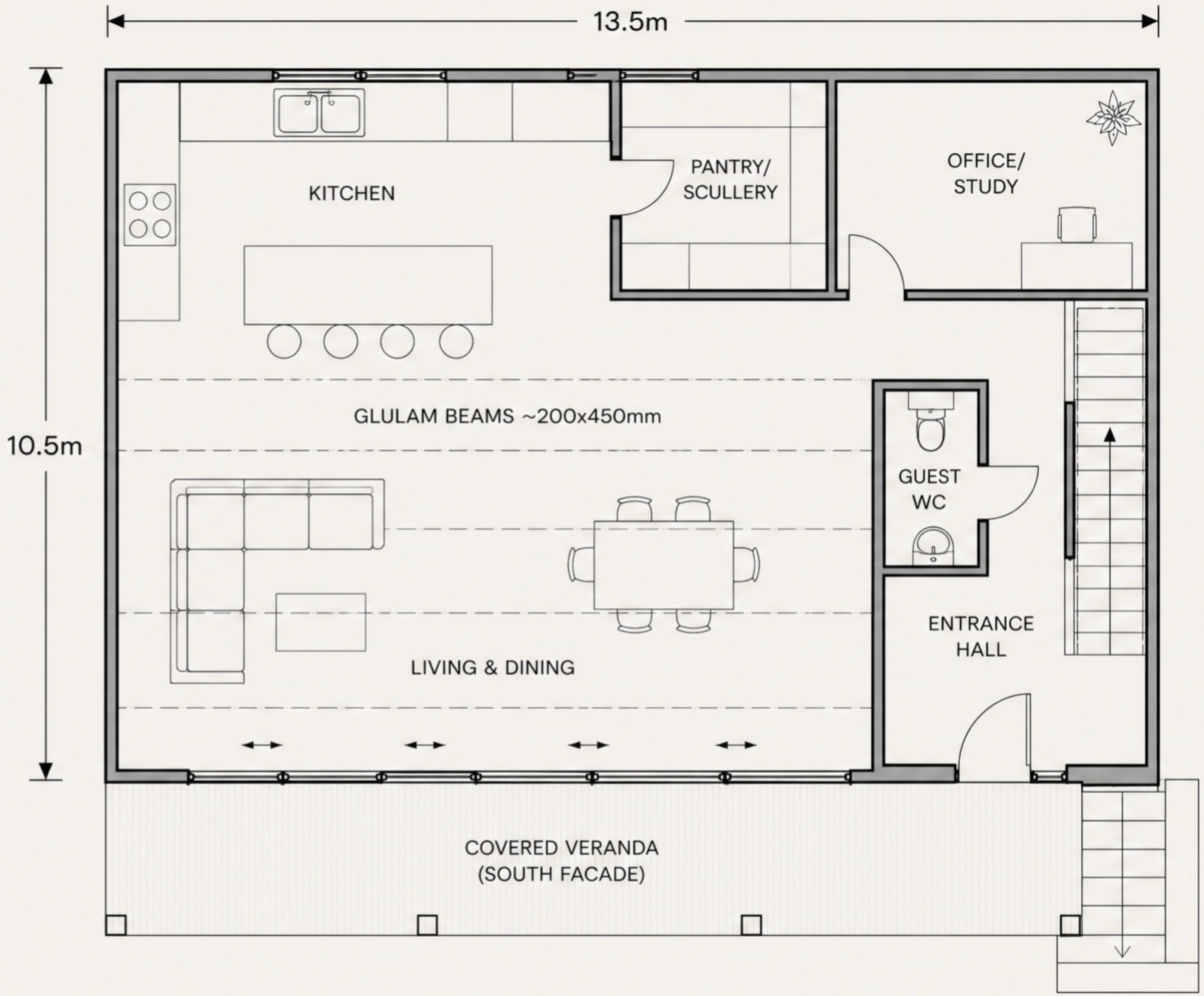


Main Living Level – Sala

Logical flow, separate service zones, and continuous glass walls

The main floor (13.5m x 10.5m) features a massive, uninterrupted sliding glass wall connecting the open-plan living area directly to the shaded veranda. The entrance and service core are logically separated: the main door opens into an entrance hall, providing access to the Guest WC and staircase. The pantry is independently accessed directly from the kitchen.

- **Logical Access:** Guest WC accessed from the entrance hall. Pantry accessed directly from the kitchen.
- **Main Entrance (East):** Entrance door located on the EAST facade, leading into a dedicated hall.
- **Continuous South Window:** Massive sliding glass wall to veranda (NO solid doors breaking the glass).
- **Office Space:** Dedicated study room (northeast corner).
- **Glulam Beams:** ~200x450mm spanning the >6m open living/kitchen area.
- **Nested Exterior Stairs:** Concrete steps on the EAST end of the south veranda.

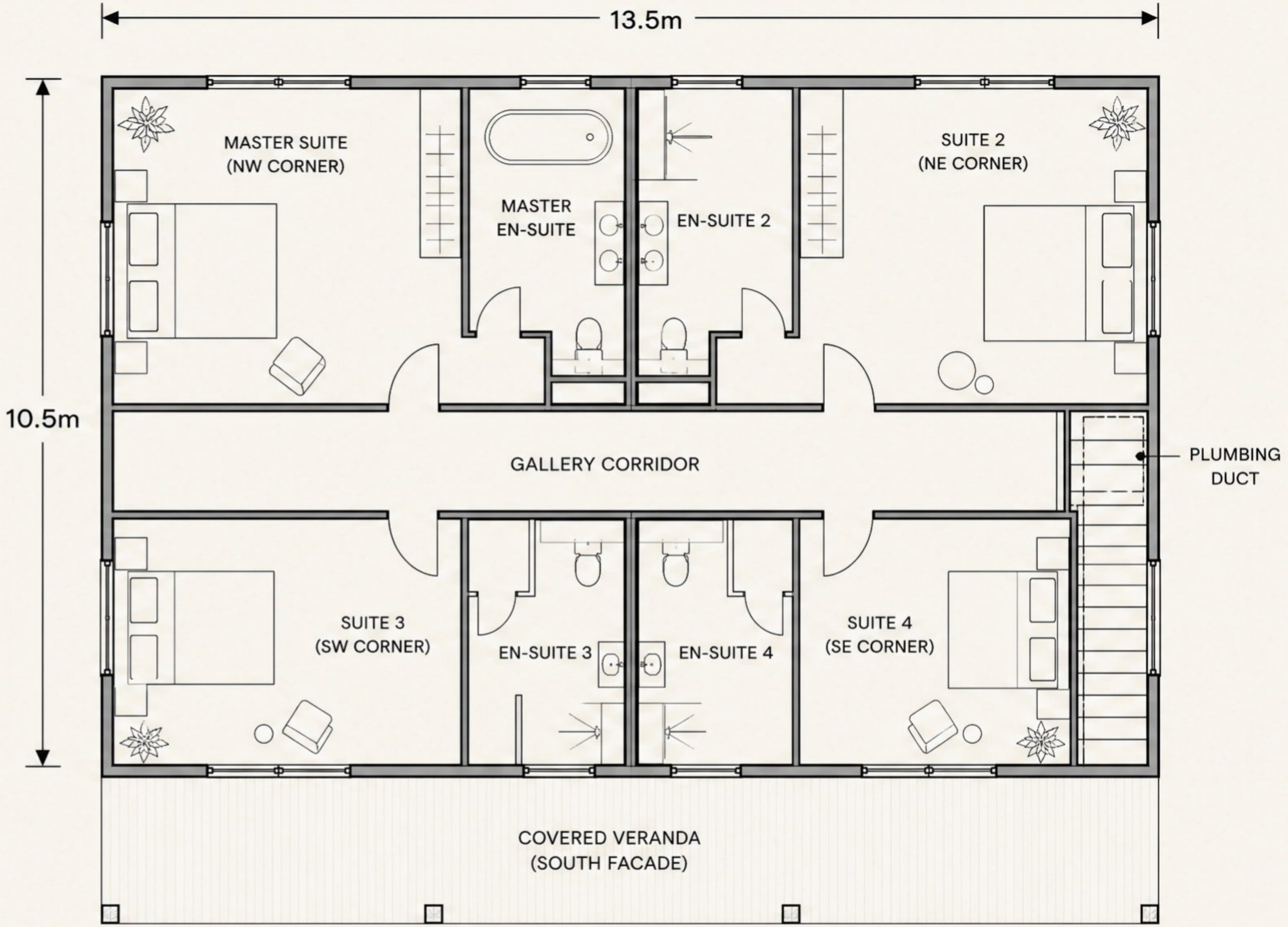


Upper Floor — Four Suites

Four symmetric suites with back-to-back plumbing and natural light

The upper floor layout is completely redesigned for architectural logic and efficiency. The 4 en-suite bathrooms are grouped in the center of the plan, placed back-to-back across the gallery corridor to share plumbing walls. This allows all 4 bedrooms to occupy the corners of the building, maximizing windows and natural light on all facades. The plumbing stack runs vertically down the east side, right next to the staircase.

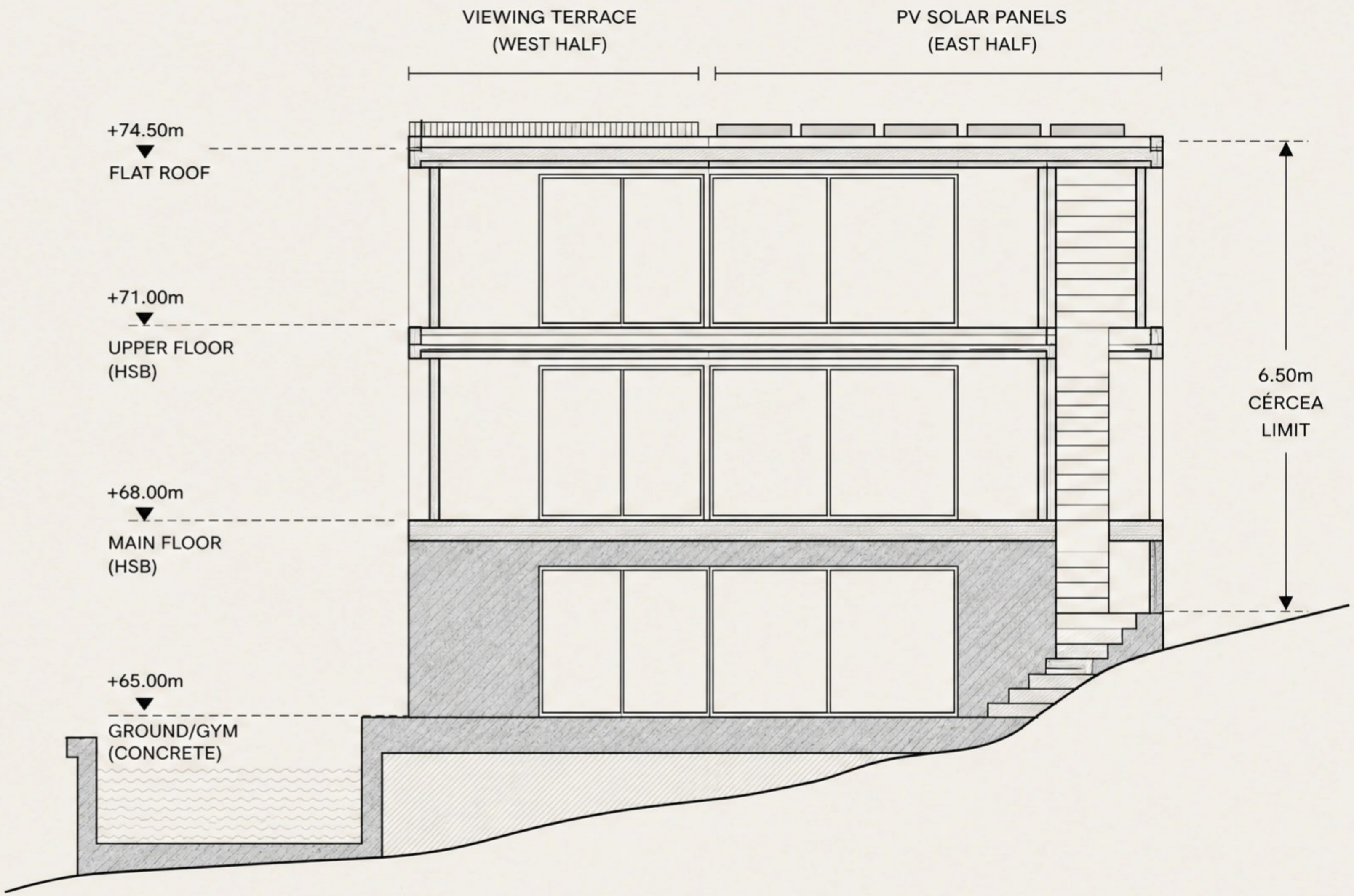
- **Central Plumbing Core:** Bathrooms placed back-to-back in the center of the plan
- **Plumbing Stack:** Runs down the EAST edge, alongside the staircase
- **Corner Bedrooms:** All 4 suites occupy the corners, with windows on TWO facades each
- **Master Suite (NW corner):** Largest suite, en-suite features a BATHTUB
- **Suites 2, 3 & 4:** En-suite pods with showers
- **Staircase:** STRICTLY ON THE EAST WALL EDGE



Accessible flat rooftop terrace and dedicated solar generation zone

The roof is a high-performance HSB cassette with absolutely zero pitch. It is divided into two functional zones: a PV solar panel array (east half) and an accessible viewing terrace (west half) offering panoramic coastal views. Total building height sits exactly at the 6.50m cércea limit.

- **Flat Roof:** Absolutely zero pitch, high-performance HSB roof cassette
- **Roof Zoning:** PV solar panels (east half) + viewing terrace (west half)
- **Side Stair Core:** Continuous vertical connection on the EAST EDGE of the plan
- **HSB Construction:** Upper two storeys entirely timber frame
- **Concrete Base:** Ground floor slab extends south to form pool terrace
- **Total height:** exactly 6.50m cércea limit



Exterior Aesthetics

A sophisticated material palette blending Algarve tradition with modern warmth

- **White Mineral Render:** Applied to the HSB panels, honoring the traditional Algarve aesthetic while providing a crisp, modern finish
- **Natural Wood Cladding:** Thermowood used inside the deep, carved verandas and soffits to introduce warmth and texture
- **Concrete Elements:** Exposed architectural concrete for the base, pool terrace, and exterior stairs, anchoring the building to the terrain
- **Slim Aluminium Frames:** Anthracite or black minimalist window profiles to maximize views and contrast with the white render

Photorealistic Concept Rendering

A refined geometric volume with elegant shaded verandas and a concrete base

The exterior aesthetic moves away from heavy monoliths to embrace extreme lightness, charm, and warmth. The traditional Algarve white render is paired with elegant, ultra-thin veranda profiles and light, natural wood cladding. A minimalist glass balustrade lines the flat roof. The ground floor concrete base is softened by a wooden pergola overhang, providing shade to the pool terrace. The home sits gracefully on a nearly flat Mediterranean landscape dotted with native pine trees.

Hybrid Construction System

Concrete foundation, timber frame walls, and targeted Glulam spans

The building employs a smart hybrid construction approach. The ground floor (cave) and pool terrace are formed from solid concrete, anchoring the house into the nearly flat terrain. The upper two storeys are built using highly insulated HSB (Houtskeletbouw) timber frame elements. Glulam (glued laminated timber) beams are used strictly and only where spans exceed 6 meters in the open-plan living area.

Concrete Base:
Ground floor and pool slab
(high thermal mass)

HSB Timber Frame:
Upper two storeys (lightweight, highly insulated, fast assembly)

Strict Glulam Usage:
Glulam beams (~200x450mm) used ONLY for open spans >6m

Flat Roof Cassette:
Highly insulated flat roof structure with glass balustrade

Veranda Shading:
Integrated timber framing for the deep, thin-profile verandas and ground floor pergola

Sustainability and Comfort

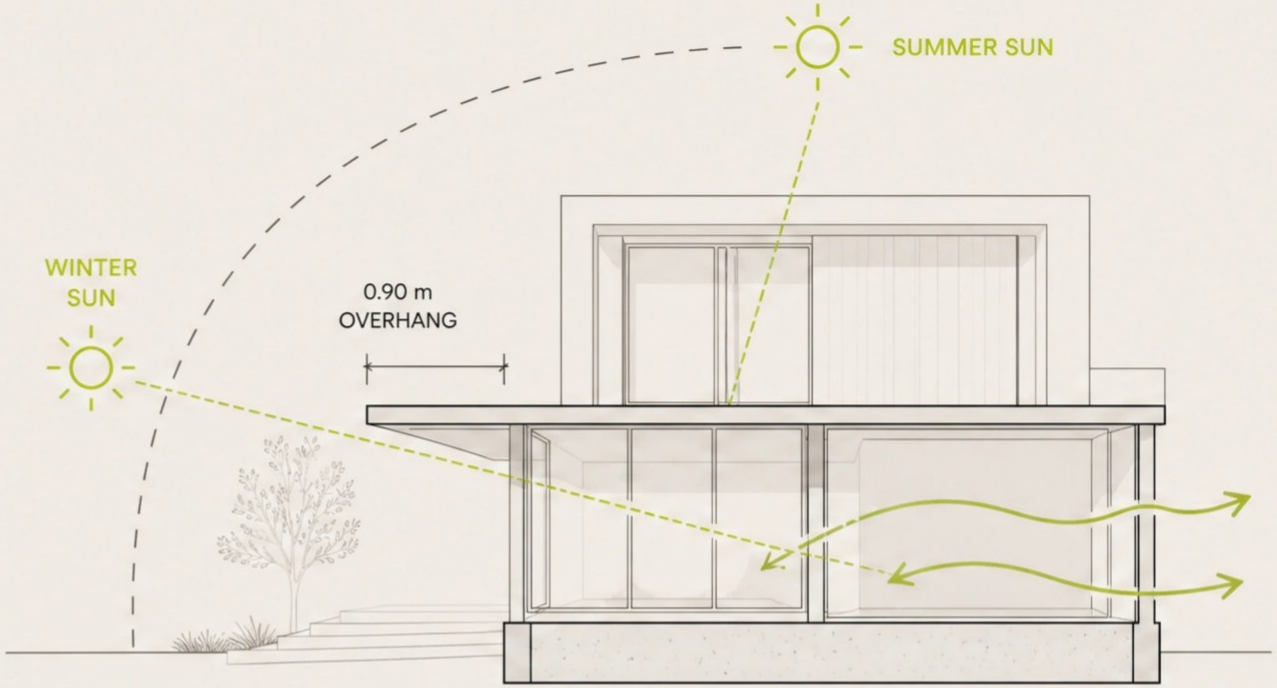
Passive comfort by design —
90% lower energy bills for life

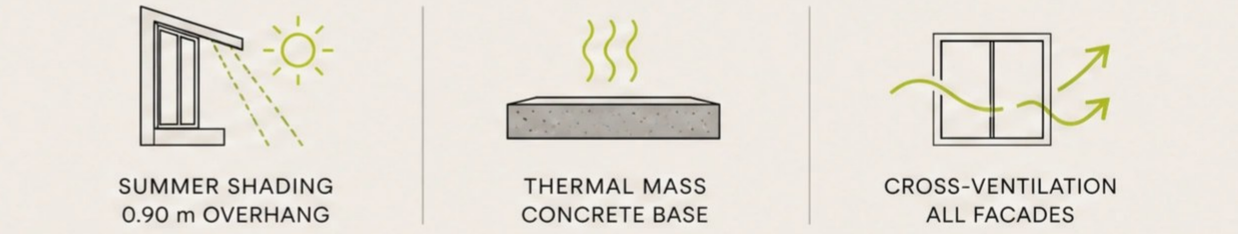
PASSIVE STRATEGIES

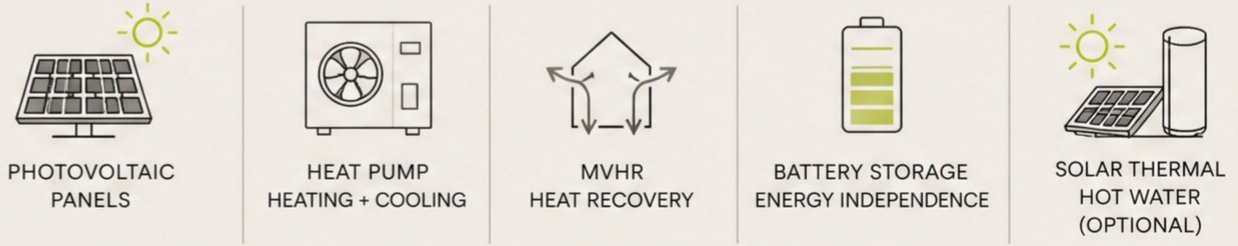
- West-facing glazing with 0.90 m overhang for summer shading
- Thermal mass at concrete base level
- Cross-ventilation through operable windows on all facades
- High-performance insulated timber frame envelope
- Low-E solar control glazing

ACTIVE SYSTEMS

- Photovoltaic gianels on rooftop
- Heat pump for heating and cooling
- MVHR (Mechanical Ventilation with Heat Recovery)
- Battery storage for energy independence
- Solar thermal for hot water (optional)







Regulatory Compliance

Fully compliant with Plano de Pormenor do Espartal

The design respects all parameters established by the local detailed plan, maintaining generous margins below maximum allowances.

Parameter	Maximum (PP Espartal)	Proposed
Implantation	Max 229.00 m²	Proposed ~142 m² (38% below max)
ABC (total area)	Max 295.00 m²	Proposed ~283 m² (Within limit)
Cércea (height)	Max 6.50 m	6.50 m (At limit)
Floors above soleira	Max 2	2 (Compliant)
Cave	Allowed	1 level (Compliant)

Closing — Next Steps

From concept to construction — next steps

This concept demonstrates a premium, sustainable residence that maximizes Lot 37's potential — its ocean views, natural slope, and southern road access — while respecting all regulatory constraints and delivering a distinctly Portuguese architectural identity.

NEXT STEPS

1. Confirm regulatory conditions with Câmara Municipal de Aljezur
2. Develop detailed floor plans and elevations
3. Structural engineering (concrete + timber + CLT coordination)
4. Advance to preliminary project (Projeto de Licenciamento)
5. MEP engineering and sustainability certification
6. Production planning and construction timeline

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